

Supplementary Material for Proximal Empirical Bayes for Sparse Regression with Posterior Decision Support

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1 Overview

This supplement gives the additional derivations, numerical diagnostics and sensitivity analyses supporting the main paper. It contains the fuller SAPG derivation, the complete Study 1 decision grid, Study 2 geometry and preconditioning diagnostics, the detailed Study 3 diabetes results, the intrinsic-dimension calibration check, and the observation-scale/Moreau sensitivity study.

2 Additional methodological details

2.1 SAPG derivation and implementation details

The main manuscript retains the score and algorithm needed to understand the empirical Bayes calculation used in the paper. Here we record the more general derivation and the implementation details. Following Fernandez Vidal et al. [2020], consider the maximum-marginal-likelihood problem

$$\hat{\theta} \in \arg \max_{\theta \in \Theta} \pi(y \mid \theta), \quad \pi(y \mid \theta) = \int \pi(y \mid \beta) \pi(\beta \mid \theta) d\beta.$$

If this objective and its gradient were available directly, a projected-gradient iteration would have the form

$$\theta_{k+1} = \Pi_{\Theta} \{ \theta_k + a_k \nabla_{\theta} \log \pi(y \mid \theta_k) \}. \quad (1)$$

The obstacle is that the marginal likelihood is generally intractable. Writing

$$Z(\theta) = \int \exp\{-\theta g(\beta)\} d\beta$$

for the prior normalising constant, Fisher’s identity gives

$$\nabla_{\theta} \log \pi(y \mid \theta) = -\mathbb{E}_{\theta}[g(\beta) \mid y] - \nabla_{\theta} \log Z(\theta). \quad (2)$$

Thus the intractable posterior expectation can be replaced by an MCMC average. For posterior states $\beta^{(1)}, \dots, \beta^{(m)}$ generated at the current value of θ ,

$$\Delta_{m,\theta} = -\nabla_{\theta} \log Z(\theta) - \frac{1}{m} \sum_{\ell=1}^m g(\beta^{(\ell)}) \quad (3)$$

is the corresponding stochastic score estimate. Substitution into (1) gives the stochastic-approximation recursion analysed in De Bortoli et al. [2020]. Under their assumptions on the stochastic-approximation schedule and the Markov kernels, the MCMC error is controlled asymptotically and tail/Polyak–Ruppert averaging stabilises the resulting estimate.

For the scalar homogeneous scale used in the present paper, the calculation simplifies substantially. If the prior is supported on a d -dimensional linear space C , let μ_d denote the induced d -dimensional Lebesgue measure on C . If $g(t\beta) = tg(\beta)$ for $t > 0$, then

$$Z_C(\theta) = \int_C e^{-\theta g(\beta)} d\mu_d(\beta) = \theta^{-d} Z_C(1).$$

With $\eta = \log \theta$, the exact score is therefore

$$\frac{\partial}{\partial \eta} \log \pi(y \mid e^{\eta}) = d - \theta \mathbb{E}_{\theta}[g(\beta) \mid y].$$

This gives the Monte Carlo quantity used in the main manuscript,

$$\widehat{\Delta}_k = d - \theta_k \bar{g}_k, \quad \bar{g}_k = \frac{1}{m_k} \sum_{\ell=1}^{m_k} g(\beta_k^{(\ell)}),$$

and the log-scale update

$$\eta_{k+1} = \Pi_{\log \Theta} \{\eta_k + \rho_k \widehat{\Delta}_k\}.$$

A tail average of the iterates is reported as $\widehat{\theta}$.

The posterior expectation entering this update is estimated by MYULA. In the unconstrained case, the inner chain is exactly the MYULA recursion introduced in the main manuscript, run at the current value θ_k . Under a hard homogeneous constraint $C_0 = \{\beta : A\beta = 0\}$, the chain is instead kept in C_0 . For the sum-zero example,

$$C_0 = \{\beta : \mathbf{1}^{\top} \beta = 0\}, \quad P_{C_0} = I - \frac{1}{p} \mathbf{1} \mathbf{1}^{\top}.$$

A constrained MYULA update can be written as

$$\begin{aligned} \beta^{(k+1)} &= \beta^{(k)} - \delta \left[P_{C_0} \nabla f_0(\beta^{(k)}) + \frac{1}{\lambda} \{ \beta^{(k)} - \text{prox}_{\lambda \theta g}^{C_0}(\beta^{(k)}) \} \right] \\ &\quad + \sqrt{2\delta} P_{C_0} \xi^{(k+1)}. \end{aligned} \quad (4)$$

Consequently, a chain initialised in C_0 remains in the subspace up to floating-point error. The constrained proximal map used in (4) is given in the following subsection.

The scale-matching identity

$$\mathbb{E}_\theta[\theta g(\beta)] = d$$

under the corresponding homogeneous prior motivates

$$\theta_0 = \frac{d}{g(\beta_{\text{ref}})}.$$

Operationally, β_{ref} is taken from a preliminary fit already available at this stage: ordinary least squares in the unconstrained and soft affine $\text{rank}(X) < n$ cases, constrained least squares under a hard homogeneous affine restriction, and the nuisance scaled Lasso coefficient vector in the full-row-rank high-dimensional branch. Unless stated otherwise, the same reference vector initialises the short MYULA chain inside SAPG. We use $d = p$ in the unconstrained and soft affine cases and $d = p - \text{rank}(A)$ under a hard homogeneous constraint.

A compact interval

$$\Theta = [\theta_0/c_\Theta, c_\Theta\theta_0]$$

prevents numerically extreme stochastic-approximation iterates. Unless otherwise stated, $c_\Theta = 10$; the Study 1 calculation and the intrinsic-dimensional verification use the wider interval reported with their numerical settings following the pilot diagnostics. In the experiments the Robbins–Monro step has the form

$$\rho_k = \frac{c}{k + k_0},$$

and a tail average is used rather than the final stochastic iterate alone. These choices concern numerical stabilisation of the marginal-likelihood search; the statistical target remains the nonsmoothed empirical Bayes objective described in the main manuscript.

2.2 Hard sum-zero proximal map

For

$$C_0 = \{\beta \in \mathbb{R}^p : \mathbf{1}^\top \beta = 0\},$$

the constrained proximal map

$$\text{prox}_{t\theta\|\cdot\|_1}^{C_0}(v) = \arg \min_{u \in C_0} \left\{ \frac{1}{2} \|u - v\|_2^2 + t\theta \|u\|_1 \right\}$$

has the KKT representation

$$u_j = S_{t\theta}(v_j - \nu), \quad \sum_{j=1}^p S_{t\theta}(v_j - \nu) = 0,$$

where $S_a(x) = \text{sign}(x)(|x| - a)_+$ is the scalar soft-thresholding operator. The scalar ν is found by bisection. If Q spans C_0 with orthonormal columns, a restricted smooth constant is

$$L_{f,C_0} = \frac{\|XQ\|_2^2}{\hat{\sigma}^2}.$$

2.3 Approximation hierarchy and smoothed-MAP audit

As in the main manuscript, f denotes the active smooth term: $f = f_0$ in the unconstrained and hard homogeneous cases, and $f = f_{\tau_c}$ in the soft affine case. The numerical hierarchy is

$$\pi_\theta \longrightarrow \pi_{\theta,\lambda} \longrightarrow \pi_{\theta,\lambda,\delta}.$$

The smoothed MAP

$$\hat{\beta}_{\text{MAP},\lambda} \in \arg \min_{\beta} \{f(\beta) + h_{\hat{\theta}}^{\lambda}(\beta)\}$$

is retained as a diagnostic. We monitor its relative L_2 displacement and its maximum coordinate shift relative to the practical threshold used in the downstream gate.

2.4 Decision-level Monte Carlo error

For indicator draws

$$I_j^{(m)} = \mathbf{1}\{|\beta_j^{(m)}| \geq \tau_{\text{post}}\},$$

the activation probability is the sample mean of $I_j^{(m)}$. Its Monte Carlo standard error (MCSE) is estimated using the effective sample size (ESS) of the indicator sequence. The decision distance

$$D_j = \frac{|\hat{\pi}_j - \pi_{\star}|}{\text{MCSE}(\hat{\pi}_j)}$$

is reported primarily over MAP-eligible variables. The MCSE entering D_j conditions on the estimated τ_{post} and therefore does not propagate Monte Carlo uncertainty in the threshold itself; the direct short-versus-long selected-set refinement audit in Study 1 provides a complementary check on the stability of the complete terminal decision.

3 Study 1: End-to-end synthetic experiment

Study 1 evaluates the complete workflow under a known sparse ground truth in two distinct dimensional regimes. The main manuscript reports the principal calibration, estimation, selection and Monte Carlo conclusions. This section records the design details, the five-dataset likelihood-scale pilot that led to the reported high-dimensional branch, the complete decision grid, signal-wise selection behaviour, posterior sample-size checks and additional numerical diagnostics. Throughout, TPR denotes true-positive rate, FDR false-discovery rate, and support RMSE the root mean squared coefficient error restricted to the true support.

3.1 Design and computational settings

Rows of the Gaussian design are generated independently from an AR(1) model,

$$X_i \sim \mathcal{N}(0, \Sigma), \quad \Sigma_{jk} = 0.4^{|j-k|},$$

and each realised column is rescaled to $\|X_j\|_2 = \sqrt{n}$. The response is

$$y = X\beta^* + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_n),$$

so that the true noise scale is $\sigma = 1$. The ten nonzero coefficients occur with indices

$$S^* = \{11, 31, 51, 71, 91, 111, 131, 151, 171, 191\},$$

with values

$$\begin{aligned} \beta_{S^*}^* = & (0.772628, -0.717440, 0.662253, -0.607065, 0.551877, \\ & -0.496690, 0.441502, -0.386314, 0.331126, -0.275939), \end{aligned}$$

obtained by rescaling the signed template $(1.4, -1.3, \dots, -0.5)$ to population signal-to-noise ratio three. The same signal vector is used in both regimes.

Table 1: Study 1 design and algorithmic settings.

Quantity	Fixed value
Replications	$R = 100$ per regime
Low-dimensional regime	$n = 500, p = 200$
High-dimensional regime	$n = 200, p = 500$
True support size	$s = 10$
AR(1) correlation	$\rho = 0.4$
True noise scale	$\sigma = 1$
Population SNR	3
Gate grid	$k \in \{2, 2.5, 3\}, \pi_* \in \{0.50, 0.75, 0.90\}$
Reference gate	$(k, \pi_*) = (2.5, 0.75)$
SAPG iterations	4000, averaging from 2000
SAPG projection factor	10^6
Initial posterior calculation	4000 retained MYULA draws after 2000 burn-in
40,000-draw posterior run	40,000 retained draws, same posterior RNG stream
MYULA rule	$\lambda = 1/L_f, c_\delta = 0.9$

An independent master seed was used for the $R = 100$ experiment after the five-dataset pilot, so none of the pilot datasets entered the reported results.

3.2 High-dimensional likelihood-scale pilot

The initial $p > n$ pilot used the raw scaled Lasso scale and a SAPG projection factor of ten. It revealed two distinct problems: the scaled Lasso scale was materially inflated, and the SAPG iterates were constrained by the projection interval. Across the five pilot datasets, the raw scaled Lasso estimate had mean $\hat{\sigma}_{\text{SL}} = 1.514$. The same datasets were then used for a controlled sequence in which one ingredient was changed at a time.

Widening the SAPG bounds removed the artificial boundary effect but did not restore selection while the inflated scale was retained. Supplying the true $\sigma = 1$ produced a much larger recovery in sensitivity and prediction performance. The feasible correction used in the reported experiment was therefore to keep the scaled Lasso only for provisional support identification, refit ordinary least squares on that support, and estimate the scale from the

Table 2: Likelihood-scale pilot on the same five high-dimensional datasets. These values are diagnostic and were not included in the reported $R = 100$ summaries.

Variant	$\hat{\sigma}$	$\hat{\theta}$	Bound hit	TPR	Size	Pred. risk
Raw SL, factor 10	1.514	34.95	1.00	0.08	0.8	1.673
Raw SL, wide bounds	1.514	15.72	0.00	0.10	1.0	0.475
True $\sigma = 1$, wide bounds	1.000	17.78	0.00	0.58	5.8	0.326
Post-OLS $\hat{\sigma}$, wide bounds	1.138	17.31	0.00	0.46	4.6	0.340

refitted residuals. In the pilot this reduced the mean estimated scale from 1.514 to 1.138 and produced behaviour much closer to the true-noise comparison.

Across the $R = 100$ datasets, the provisional scaled Lasso support has mean size 7.72, contains on average 7.33 of the ten true variables, and contains 0.39 null variables. Sixty-six provisional supports contain no false positives, 29 contain one and five contain two. Hence the refit removes the direct ℓ_1 -shrinkage contribution to the residuals, but weak signals omitted by the provisional support can still inflate the resulting scale estimate.

Table 3: Noise-scale distributions across the reported $R = 100$ experiment.

Estimator	Mean	SD	Median	Min	Max
Residual, $n > p$	0.9979	0.0415	0.9924	0.9133	1.0922
Raw scaled Lasso, $p > n$	1.4820	0.0796	1.4832	1.3205	1.6686
Post-OLS, $p > n$	1.1173	0.0776	1.1147	0.9622	1.3096

Figure 1 complements the numerical summaries in Table 3 by showing the full empirical distributions of the three scale estimates. The low-dimensional residual estimator is centred close to the true value $\sigma = 1$, whereas the raw scaled Lasso estimate is shifted upward throughout the high-dimensional experiment. The post-selection OLS refit removes a substantial part of this shift, although a smaller positive bias remains.

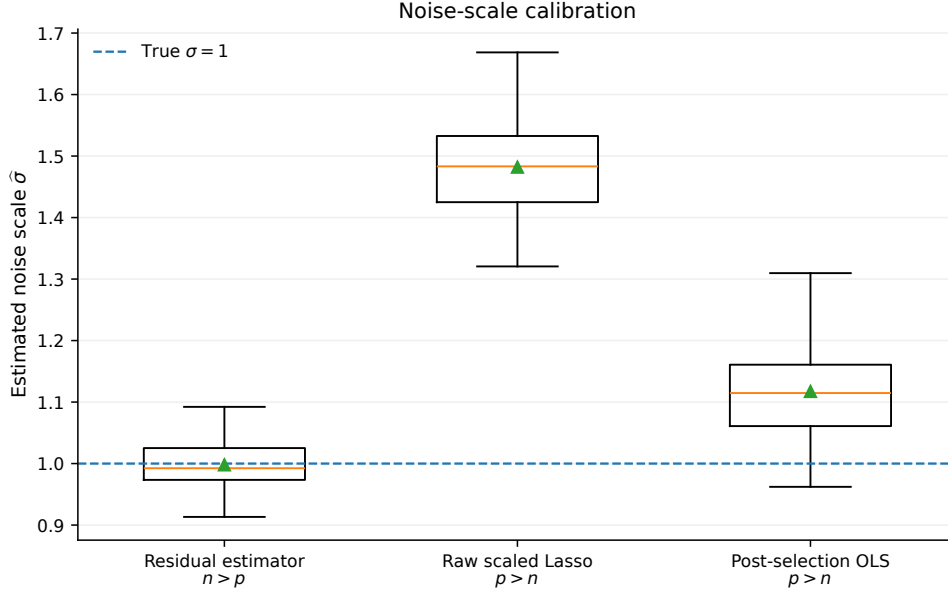


Figure 1: Noise-scale calibration across the $R = 100$ Study 1 datasets. The dashed horizontal line marks the true noise scale $\sigma = 1$.

3.3 Reference performance and support-error observations

Table 4: Study 1 performance at the reference gate $(k, \pi_\star) = (2.5, 0.75)$. Parentheses contain standard errors across the 100 independently generated datasets.

Regime	$\hat{\sigma}/\sigma$	$\hat{\theta}$	Rel. L_2	Supp. RMSE	Pred. risk	TPR	FDR	Size	Exact
500, 200	0.998 (.004)	15.711 (.041)	0.215 (.003)	0.065 (.001)	0.130 (.003)	0.989 (.003)	0.006 (.002)	9.96 (.04)	0.830 (.038)
200, 500	1.117 (.008)	17.334 (.067)	0.343 (.004)	0.165 (.003)	0.347 (.009)	0.429 (.012)	0.000 (.000)	4.29 (.12)	0.000 (.000)

At the reference gate, the low-dimensional selected-set size equals nine in ten datasets, ten in 84, and eleven in six. Exact support is recovered in 83 datasets. Of the remaining 17, six contain one false positive and no false negative, ten contain one false negative and no false positive, and one contains both one false positive and one false negative.

In the high-dimensional regime, every coordinate selected at the reference gate is a true signal. Selected-set sizes range from one to eight, with median four:

$ \hat{S} $	1	2	3	4	5	6	7	8
frequency	1	6	18	32	27	14	1	1.

Thus the failure of exact support recovery in this regime is entirely due to false negatives.

3.4 Complete decision grid

The grid below is calculated by post-processing the same posterior sample in each replicate; $\hat{\sigma}^2$, $\hat{\theta}$, the MAP and the MYULA draws are unchanged across gate settings.

Table 5: Study 1 decision grid using 40,000 retained MYULA draws.

Regime	(k, π_*)	TPR	FDR	Size	Exact
500, 200	(2, .50)	1.000	0.1047	11.33	0.37
	(2, .75)	0.996	0.0161	10.14	0.81
	(2, .90)	0.988	0.0018	9.90	0.86
	(2.5, .50)	1.000	0.0314	10.36	0.72
	(2.5, .75)	0.989	0.0065	9.96	0.83
	(2.5, .90)	0.972	0.0000	9.72	0.74
	(3, .50)	0.994	0.0083	10.03	0.86
	(3, .75)	0.981	0.0000	9.81	0.81
	(3, .90)	0.949	0.0000	9.49	0.56
200, 500	(2, .50)	0.675	0.0065	6.80	0
	(2, .75)	0.490	0.0000	4.90	0
	(2, .90)	0.340	0.0000	3.40	0
	(2.5, .50)	0.595	0.0000	5.95	0
	(2.5, .75)	0.429	0.0000	4.29	0
	(2.5, .90)	0.283	0.0000	2.83	0
	(3, .50)	0.520	0.0000	5.20	0
	(3, .75)	0.364	0.0000	3.64	0
	(3, .90)	0.229	0.0000	2.29	0

3.5 Signal-wise eligibility and selection

The monotone decline in high-dimensional sensitivity is easier to interpret at the level of the ten true coordinates. “Eligible” denotes the fraction of replicates in which the nonsmoothed MAP clears the practical magnitude threshold before application of the posterior activation-probability gate.

Table 6: Reference-gate eligibility and selection frequency for each true signal.

Index	β_j^*	$n = 500, p = 200$		$n = 200, p = 500$	
		Eligible	Selected	Eligible	Selected
11	.773	1.00	1.00	1.00	.96
31	-.717	1.00	1.00	1.00	.94
51	.662	1.00	1.00	1.00	.85
71	-.607	1.00	1.00	1.00	.67
91	.552	1.00	1.00	1.00	.51
111	-.497	1.00	1.00	.99	.20
131	.442	1.00	1.00	.94	.11
151	-.386	1.00	1.00	.87	.04
171	.331	1.00	.99	.59	.01
191	-.276	1.00	.90	.33	.00

Table 6 gives the exact coordinatewise frequencies, while the corresponding high-dimensional

profile is shown in Figure 1 of the main manuscript. MAP eligibility remains essentially certain for the stronger and moderate signals, whereas the final selection frequency declines steadily with signal magnitude: for the first five signals it falls from 0.96 to 0.51 even though all five are MAP-eligible in every replicate. For the weaker signals, MAP eligibility also begins to decline, and the posterior gate becomes increasingly restrictive. Across all $100 \times (500 - 10) = 49,000$ high-dimensional null-coordinate opportunities, no null coordinate is selected at the reference setting.

3.6 Posterior refinement and decision-level Monte Carlo error

The initial posterior run retained 4000 draws. A second run increased this to 40,000 retained draws while keeping the data, $\hat{\sigma}^2$, $\hat{\theta}$, MAP, Moreau parameter and timestep fixed. The same posterior random-number seeds were used, so that the 4000-draw calculation is a prefix of the longer run.

Table 7: Selection stability between the 4000- and 40,000-draw MYULA calculations.

Regime	TPR _{4k}	TPR _{40k}	FDR _{4k}	FDR _{40k}	Sets identical	Mean set difference
500, 200	.990	.989	.0036	.0065	.96	.04
200, 500	.424	.429	.0000	.0000	.74	.33

Table 8: Monte Carlo precision for the 4000- and 40,000-draw MYULA calculations. Here “smallest eligible D_j ” denotes, within each replicate, the smallest decision-distance diagnostic among coordinates passing the MAP magnitude gate.

Regime	Med. ESS 4k	Med. ESS 40k	Med. smallest eligible D_j 4k	Med. smallest eligible D_j 40k
500, 200	212	2093	7.03	27.87
200, 500	134	1269	.81	2.16

For each replicate, let $\hat{S}_{4k}$ and $\hat{S}_{40k}$ denote the selected sets obtained from the 4000- and 40,000-draw calculations. The “mean set difference” in Table 7 is based on the symmetric-difference size

$$|\hat{S}_{4k} \triangle \hat{S}_{40k}|,$$

which counts coordinates selected in exactly one of the two runs. Its empirical distribution is

	0	1	2	3
$n = 500, p = 200$	96	4	0	0
$n = 200, p = 500$	74	20	5	1.

Figure 2 displays the same selected-set comparison at the replicate level. The two calculations give identical selected sets in 96% of the low-dimensional datasets and 74% of the high-dimensional datasets; when a difference occurs, it usually consists of a single coordinate selected in exactly one of the two runs.

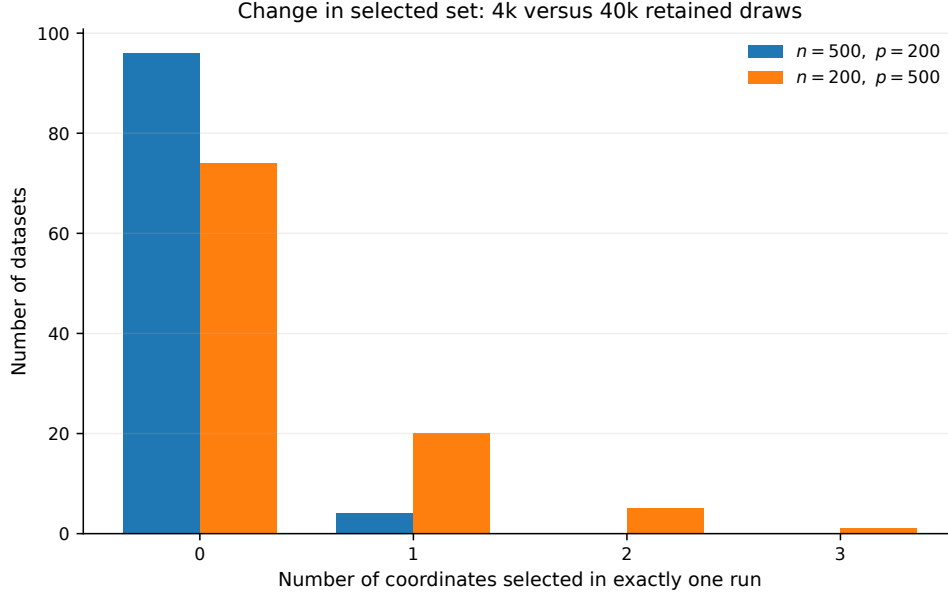


Figure 2: Change in the selected set when the number of retained MYULA draws is increased from 4000 to 40,000. The horizontal categories count coordinates selected in exactly one of the two calculations.

The minimum-coordinate ESS in the 40,000-draw runs has median approximately 976 in the low-dimensional regime and 263 in the high-dimensional regime; the worst observed values are 630 and 138, respectively. For the decision diagnostic, we take within each replicate the smallest D_j among coordinates that pass the MAP magnitude gate. This identifies the eligible coordinate whose estimated activation probability is closest to π_* in Monte Carlo standard-error units. The median of this quantity after 40,000 draws is 27.87 for $n > p$ and 2.16 for $p > n$. Only 5% of low-dimensional replicates, but 46% of high-dimensional replicates, contain an eligible coordinate within two MCSE units of the activation threshold. The longer run therefore leaves aggregate TPR and FDR essentially unchanged while materially improving the Monte Carlo precision of the individual decisions most likely to change near the gate.

Figure 2 of the main manuscript shows this change in decision-level Monte Carlo precision directly. The low-dimensional calculations are already well separated from the probability boundary after 4000 retained draws, whereas in the high-dimensional regime the longer run moves the distribution of the smallest eligible D_j away from that boundary and reduces the number of replicates for which an individual decision remains Monte Carlo-sensitive.

3.7 Exact-versus-smoothed MAP audit

Write

$$\Delta_{\text{MAP}} = \hat{\beta}_{\text{MAP},\lambda} - \hat{\beta}_{\text{MAP}}, \quad \tau_{\text{ref}} = \tau_{\text{post}}(\lambda; 2.5).$$

For this audit, the threshold-normalised maximum shift is $\|\Delta_{\text{MAP}}\|_{\infty}/\tau_{\text{ref}}$, while the relative L_2 shift is $\|\Delta_{\text{MAP}}\|_2/\|\hat{\beta}_{\text{MAP}}\|_2$. These normalisations are specific to the supplementary

comparison and are distinct from the generic $D_2(\lambda)$ diagnostic in the main manuscript.

Table 9: Original-to-smoothed MAP discrepancy in units of the reference posterior threshold.

Regime	Median $\ \Delta_{\text{MAP}}\ _{\infty}/\tau_{\text{ref}}$	Maximum	Mean relative L_2 shift	Maximum relative L_2 shift
$n = 500, p = 200$	0.068	0.110	0.0308	0.0399
$n = 200, p = 500$	0.234	0.367	0.1200	0.1998

The discrepancy is larger in $p > n$, where the likelihood geometry is weaker relative to the prior, but the maximum coordinate displacement remains below one reference threshold in every replicate. This audit concerns the approximation used for posterior computation; the deterministic eligibility gate itself uses the MAP of the original nonsmoothed objective.

3.8 Detailed high-dimensional replicate

High-dimensional replicate zero was prespecified for detailed storage before inspection of the $R = 100$ results. At the reference gate, its true-coordinate summaries are:

Table 10: True coordinates in the prespecified high-dimensional replicate zero.

Index	β^*	MAP	Post. mean	Activation prob.	Selected
11	0.773	0.591	0.451	0.986	Yes
31	-0.717	-0.566	-0.407	0.966	Yes
51	0.662	0.509	0.268	0.787	Yes
71	-0.607	-0.535	-0.255	0.753	Yes
91	0.552	0.388	0.217	0.638	No
111	-0.497	-0.290	-0.195	0.569	No
131	0.442	0.244	0.100	0.227	No
151	-0.386	-0.301	-0.167	0.474	No
171	0.331	0.099	0.060	0.097	No
191	-0.276	-0.177	-0.080	0.153	No

Signal 171 does not clear the MAP magnitude threshold in this replicate; the remaining nine true coordinates do. Among those eligible coordinates, the posterior gate retains the four strongest and rejects the remaining five at $\pi_{\star} = 0.75$. The replicate therefore gives a concrete example of how the two gates, the sparse MAP screen and the posterior activation one, act sequentially.

3.9 Additional computational diagnostics

Table 11: Selected computational diagnostics for the final Study 1 analysis.

Regime	Med. $\ \Delta_{\text{MAP}}\ _{\infty}/\tau_{\text{ref}}$	Max. $\ \Delta_{\text{MAP}}\ _{\infty}/\tau_{\text{ref}}$	Med. ESS	Min. ESS	Med. min. D_j (eligible)	Replicates min. $D_j < 2$	Max SAPG bound hit	Median core time (s)
500, 200	0.068	0.110	2093	630	27.87	5%	0.000	7.14
200, 500	0.234	0.367	1269	138	2.16	46%	0.000	8.29

The timings include data generation, likelihood-scale estimation, SAPG, the original and smoothed MAP calculations and the 40,000-draw MYULA run, but exclude exhaustive post-processing diagnostics. They are environment-dependent and are included only as an order-of-magnitude computational record.

4 Study 2: Soft affine information

Study 2 is designed to separate the inferential and numerical consequences of a correctly specified soft affine relation. The main manuscript reports the principal sweep and the geometry-aware correction at the strongest affine setting. Here we record the exact simulation construction, additional posterior summaries, the baseline scalar-step geometry diagnostic, the complete decision grid and the timestep audit used for the retained $\kappa_c = 10$ calculation.

4.1 Design and affine-strength parameterisation

We use $R = 100$ independently generated datasets with $n = 200$, $p = 100$ and known $\sigma = 1$. The true coefficient vector is

$$\beta^* = (1, 0.8, 0.6, 0.4, -0.8, -0.5, -0.3, -0.2, 0, \dots, 0)^\top,$$

so that the support size is eight,

$$\mathbf{1}^\top \beta^* = 1, \quad \|\beta^*\|_1 = 4.6.$$

Each design matrix is generated with independent standard-Gaussian entries, and its realised columns are rescaled to Euclidean norm $\sqrt{n}$.

For $\kappa_c > 0$, the smooth potential contains the pseudo-observation term

$$\frac{1}{2\tau_c^2}(\mathbf{1}^\top \beta - 1)^2,$$

while $\kappa_c = 0$ denotes the unconstrained comparison. Let

$$L_0 = \lambda_{\max}(X^\top X), \quad \kappa_c = \frac{p}{\tau_c^2 L_0}.$$

The additional Hessian is

$$H_c = \tau_c^{-2} \mathbf{1} \mathbf{1}^\top,$$

which has rank one and nonzero eigenvalue

$$\lambda_{\max}(H_c) = \frac{p}{\tau_c^2} = \kappa_c L_0.$$

Thus, as noted in the main text, κ_c is a directly interpretable curvature ratio rather than merely a reparameterisation of τ_c . We use

$$\kappa_c \in \{0, 0.1, 1, 10\}.$$

Because the sum-to-one information is included as a Gaussian pseudo-observation rather than by restricting the prior support, the Laplace prior remains defined on $\mathbb{R}^p$ and the SAPG score uses the ambient dimension $d = p = 100$ at every κ_c . This preserves the simple homogeneous-prior calibration while allowing the nonhomogeneous affine relation to enter through the likelihood geometry.

The decision rule is evaluated over

$$k \in \{2, 2.5, 3\}, \quad \pi_\star \in \{0.5, 0.75, 0.9\},$$

with (2.5, 0.75) as the reference setting. For $\kappa_c \leq 1$, the final posterior calculation uses the scalar-step MYULA rule with the default Moreau choice $\lambda = 1/L_f$. At $\kappa_c = 10$, that default calculation is retained as a geometry diagnostic, while the final reported posterior uses the rank-one preconditioning matrix and tangent-curvature Moreau scale described below. The retained geometry-aware calculation uses the timestep multiplier $c_\delta = 0.45$. The empirical Bayes estimate and nonsmoothed MAP are common to the two $\kappa_c = 10$ posterior calculations.

4.2 Effect of affine-information strength

Table 12 records the principal coefficient-level effects. The affine information produces a large monotone improvement in adherence to the true sum-to-one relation, while the changes in shrinkage and estimation are comparatively modest. The posterior mean and the gated estimators show the same pattern.

At the reference decision setting, the “gated MAP” and “gated posterior mean” are obtained by retaining the corresponding coefficient values on $\hat{S}_\lambda(2.5, 0.75)$ and setting all remaining coordinates to zero.

Table 12: Study 2 coefficient-estimation and affine-adherence summaries. The $\kappa_c = 10$ row uses the retained geometry-aware posterior calculation.

κ_c	$\hat{\theta}$	MAP rel. L_2	Post. mean rel. L_2	Gated MAP rel. L_2	Gated post. mean rel. L_2	$E \mathbf{1}^\top \beta - 1 $
0	9.772323	.255660	.321033	.220664	.228289	.685060
.1	9.785377	.254890	.320225	.218619	.226449	.595477
1	9.975943	.251763	.316081	.213842	.222001	.341807
10	10.346372	.248190	.312500	.212324	.221139	.111777

At the reference gate, the corresponding sparse-decision and Monte Carlo summaries are shown below. Weak-tangent ESS is computed after projecting the chain onto the least-curved eigendirection of the smooth Hessian within the tangent space orthogonal to $\mathbf{1}$; it is a directional diagnostic for the part of the posterior most liable to move slowly once the scalar timestep is controlled by the affine direction.

Table 13: Study 2 reference-gate selection and posterior-computation summaries. For $\kappa_c = 10$, the row again uses the retained geometry-aware posterior calculation. D_{elig} is the median, across datasets, of the smallest decision distance among MAP-eligible coordinates.

κ_c	Mean size	TPR	FDR	Exact	Med. ESS	Weak-tangent ESS	D_{elig}
0	6.44	.80250	.002679	.05	301.6	146.5	5.49
.1	6.47	.80625	.002679	.05	302.9	146.9	5.49
1	6.53	.81500	.001250	.07	212.7	106.5	4.31
10	6.58	.82000	.002500	.07	302.7	160.8	5.03

4.3 Low-rank stiffness at $\kappa_c = 10$

The baseline scalar-step $\kappa_c = 10$ calculation makes the mechanism especially clear. Across the $R = 100$ datasets,

$$\text{mean}\left(\frac{L_f}{L_\perp}\right) = 10.3977,$$

and the mean squared alignment of the leading smooth-Hessian eigenvector with the normalised sum direction

$$u = \frac{\mathbf{1}}{\sqrt{p}}$$

is 0.999366. Equivalently, if v_1 denotes the unit leading eigenvector of the smooth Hessian, this statistic is the ensemble mean of $|v_1^\top u|^2$, the squared cosine of the angle between the leading-curvature and sum directions. A value this close to one means that the two directions are almost parallel. Thus, almost all of the largest curvature is concentrated in the one-dimensional sum direction.

The resulting scalar-step chain is not uniformly inefficient. The sum direction itself has median ESS 1107.7, whereas median coordinate ESS is only 33.8, median minimum coordinate ESS is 9.1, and weak-tangent ESS is 20.1. The stiff affine direction therefore dictates the admissible timestep even though the main mixing loss occurs away from that direction.

We use the projectors

$$P_\parallel = uu^\top, \quad P_\perp = I - P_\parallel,$$

and the rank-one preconditioning matrix

$$M_\phi = P_\perp + \phi P_\parallel, \quad \phi = \min\{1, L_\perp/L_f\}.$$

The mean value of ϕ is 0.096179. With

$$L_{f,M} = \lambda_{\max}\left(M_\phi^{1/2} H_f M_\phi^{1/2}\right)$$

denoting the smooth-gradient Lipschitz constant in the preconditioned geometry, the mean transformed smooth-curvature ratio is

$$\frac{L_{f,M}}{L_\perp} = 1.02217.$$

We also set the Moreau parameter from $L_\perp$, rather than the affine-dominated L_f . If λ were held fixed, introducing M_ϕ would change only the geometry of the Langevin dynamics and would leave the intended smoothed posterior $\pi_{\hat{\theta},\lambda}$ unchanged. Here we additionally replace the default $\lambda = 1/L_f$ by $\lambda = 1/L_\perp$. Since $L_\perp < L_f$, this gives a larger Moreau parameter, so the smoothed approximation itself also changes. Across the ensemble, this makes the tangent-based Moreau scale 10.3977 times the default $\kappa_c = 10$ value. The retained branch should therefore be interpreted as a geometry-aware posterior calculation rather than as a pure preconditioner-only experiment.

Table 14: Baseline scalar-step and retained geometry-aware $\kappa_c = 10$ posterior calculations at the reference decision setting. The empirical Bayes estimate and nonsmoothed MAP are common to both branches.

Branch	Med. SD	Med. ESS	Min. ESS	Weak-tangent ESS	Sum-dir. ESS	$E[\mathbf{1}^\top \beta - 1]$	Size	TPR	FDR	D_{elig}
Baseline scalar-step	.06341	33.8	9.1	20.1	1107.7	.11965	6.65	.825	.00625	1.25
Geometry-aware	.06616	302.7	127.4	160.8	965.4	.11178	6.58	.820	.00250	5.03

The statistical summaries are close, but the Monte Carlo diagnostics are not. In particular, the reference selected-set size and TPR change only slightly. For each dataset, we take the MAP-eligible coordinate whose estimated activation probability is closest to $\pi_\star$, and measure that separation in activation-probability MCSEs. The median of this quantity across datasets rises from 1.25 to 5.03, showing that the probability-gate decisions become much more clearly separated from their threshold. This is the distinction emphasised in the main paper: the affine information changes the posterior, but the severe loss of scalar-step efficiency at $\kappa_c = 10$ is a numerical consequence of its low-rank curvature.

Figure 3 gives a compact visual summary of these two effects. The baseline scalar-step MYULA calculation—using no geometry-aware preconditioning and a single Euclidean timestep controlled by the global smoothness constant L_f —is followed across the full κ_c sequence, with the retained geometry-aware calculation shown separately at $\kappa_c = 10$. The left panel shows that stronger affine information already produces a substantial improvement in sum-to-one adherence before the geometry correction is applied; the correction changes that posterior summary only modestly further. The right panel, by contrast, shows the pronounced collapse in coordinatewise sampling efficiency under the baseline calculation at $\kappa_c = 10$, and its recovery under the geometry-aware calculation. Thus, the large numerical gain is accompanied by only a comparatively small additional change in affine adherence.

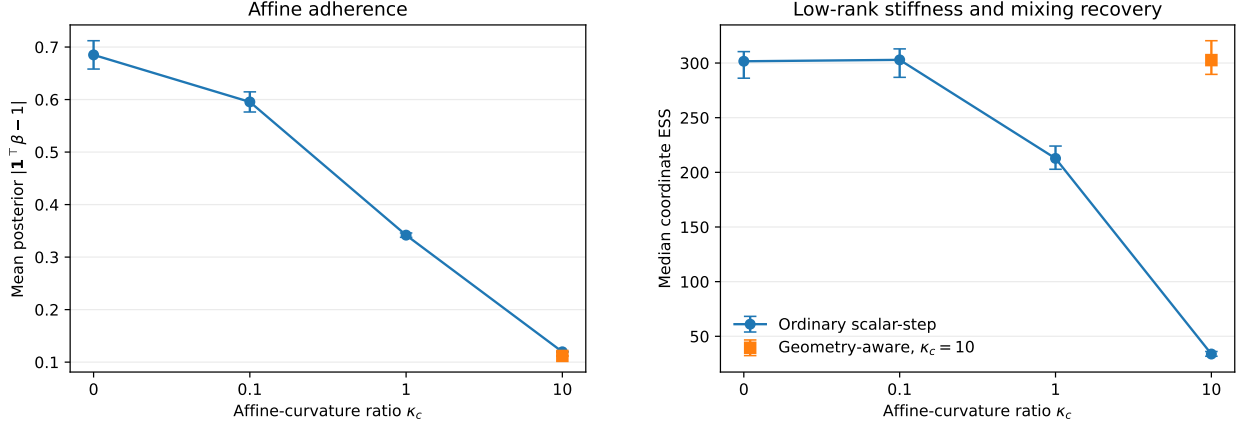


Figure 3: Study 2: affine adherence (left) and median coordinate ESS (right) as κ_c increases. The blue curve shows the baseline scalar-step MYULA calculation, while the separate marker at $\kappa_c = 10$ shows the retained geometry-aware calculation. Bars show 1.96 standard errors across datasets on the left and interquartile ranges on the right.

4.4 Complete decision grid

Table 15 reports the complete gate grid using the final posterior calculation for each κ_c , including the geometry-aware branch at $\kappa_c = 10$. The usual sensitivity–specificity pattern persists throughout: decreasing either threshold increases recovery but also permits more false discoveries. No qualitatively different support behaviour emerges at the strongest affine setting after the geometry correction.

Table 15: Study 2 complete decision grid. “Exact” is the fraction of the $R = 100$ datasets in which the selected support equals the eight-variable true support.

κ_c	k	π_*	Mean size	TPR	FDR	Exact
0	2	0.50	7.94	0.90375	0.078858	0.16
0	2	0.75	6.79	0.83750	0.011607	0.09
0	2	0.90	6.34	0.79125	0.001250	0.04
0	2.5	0.50	7.26	0.87750	0.029302	0.14
0	2.5	0.75	6.44	0.80250	0.002679	0.05
0	2.5	0.90	5.98	0.74625	0.001429	0.01
0	3	0.50	6.66	0.82625	0.006429	0.08
0	3	0.75	6.20	0.77375	0.001429	0.02
0	3	0.90	5.66	0.70750	0.000000	0.01
0.1	2	0.50	7.94	0.90375	0.077738	0.17
0.1	2	0.75	6.80	0.83875	0.011607	0.09
0.1	2	0.90	6.35	0.79250	0.001250	0.05
0.1	2.5	0.50	7.27	0.87750	0.030730	0.14

Continued on next page

Table 15 — continued

κ_c	k	π_*	Mean size	TPR	FDR	Exact
0.1	2.5	0.75	6.47	0.80625	0.002679	0.05
0.1	2.5	0.90	5.99	0.74750	0.001429	0.01
0.1	3	0.50	6.66	0.82625	0.006429	0.08
0.1	3	0.75	6.22	0.77625	0.001429	0.02
0.1	3	0.90	5.67	0.70875	0.000000	0.01
1	2	0.50	8.02	0.90750	0.082309	0.14
1	2	0.75	6.88	0.84875	0.011429	0.10
1	2	0.90	6.38	0.79625	0.001250	0.05
1	2.5	0.50	7.29	0.88000	0.030552	0.16
1	2.5	0.75	6.53	0.81500	0.001250	0.07
1	2.5	0.90	6.08	0.75875	0.001429	0.02
1	3	0.50	6.71	0.83375	0.005179	0.09
1	3	0.75	6.27	0.78250	0.001250	0.02
1	3	0.90	5.74	0.71750	0.000000	0.01
10	2	0.50	8.09	0.91250	0.085297	0.15
10	2	0.75	6.96	0.85625	0.013611	0.10
10	2	0.90	6.40	0.79750	0.002679	0.05
10	2.5	0.50	7.34	0.88375	0.033052	0.19
10	2.5	0.75	6.58	0.82000	0.002500	0.07
10	2.5	0.90	6.12	0.76375	0.001250	0.02
10	3	0.50	6.77	0.84000	0.006429	0.09
10	3	0.75	6.30	0.78625	0.001250	0.04
10	3	0.90	5.79	0.72375	0.000000	0.01

4.5 Timestep audit for the geometry-aware $\kappa_c = 10$ calculation

Changing from the default scalar calculation to the geometry-aware branch also changes the numerical scale of the discretisation. We therefore repeated the geometry-aware $\kappa_c = 10$ posterior calculation at three values of the timestep multiplier c_δ , increasing the number of transitions as required so that the total simulated Langevin time, approximately $T = N_{\text{step}}\delta$, was held fixed. Table 16 shows a small but systematic decrease in several posterior summaries as the timestep is reduced. The ESS values remain of the same order because the comparison is made at approximately matched total Langevin time rather than matched transition count. These are separate matched-time audit runs rather than the production posterior run used in Tables 12–13; consequently, the $c_\delta = 0.45$ Monte Carlo summaries need not coincide exactly with the retained-row summaries above.

The monotone movement indicates a visible finite-step discretisation effect at $c_\delta = 0.9$. We retain $c_\delta = 0.45$ as the reported $\kappa_c = 10$ setting: it materially reduces that effect without sacrificing the mixing gains delivered by the geometry correction. The $c_\delta = 0.225$ calculation is retained as a further audit rather than as the primary branch.

Table 16: Geometry-aware $\kappa_c = 10$ timestep audit at approximately matched total Langevin time $T = N_{\text{step}}\delta$.

c_δ	Med. ESS	Weak-tangent ESS	Med. SD	$E \mathbf{1}^\top \beta - 1 $	$E\ \beta\ _1$
.900	310.2	161.1	.06865	.11853	10.3040
.450	296.9	158.8	.06636	.11182	10.1162
.225	291.7	159.8	.06533	.10689	10.0265

5 Study 3: Diabetes details

Study 3 is the real-data application of the complete workflow. The main manuscript retains the study aim, the frozen calibration and posterior adequacy at summary level, the full decision-support grid, and the external interval comparison. This section records the underlying geometry, calibration, MAP and MYULA audits, coefficient-level gate diagnostics, and complete benchmark numbers.

5.1 Data, scaling and frozen calibration

The diabetes data are analysed in the same operational scaling as the public notebook accompanying Zhou et al. [2024]. The response is centred and divided by its sample standard deviation, while the supplied predictor matrix is used directly; its columns are centred and have Euclidean norm essentially equal to one. The design has $n = 442$, $p = 10$ and full column rank. The smallest and largest eigenvalues of $X^\top X$ are 0.00856053 and 4.02421418, giving a condition number of $\kappa(X^\top X) = 470.09$. The largest absolute pairwise predictor correlation is 0.896663, attained by S1 and S2.

Because an intercept is fitted implicitly through centring, the classical residual-variance calculation uses one additional degree of freedom,

$$\hat{\sigma}^2 = \frac{\|y - X\hat{\beta}_{\text{OLS}}\|_2^2}{n - p - 1} = 0.49343968, \quad \hat{\sigma} = 0.70245262.$$

This fixes

$$L_f = \frac{\lambda_{\max}(X^\top X)}{\hat{\sigma}^2} = 8.15543282, \quad \lambda = L_f^{-1} = 0.12261765,$$

and the scalar MYULA timestep

$$\delta = \frac{0.9}{L_f + 1/\lambda} = 0.05517794.$$

The natural SAPG initial value is

$$\theta_0 = \frac{p}{\|\hat{\beta}_{\text{OLS}}\|_1} = 0.22281184.$$

The SAPG calculation was extended to 60,000 iterations because the additional cost was negligible. Averaging from iteration 20,000 onward gives

$$\hat{\theta} = 0.33728856.$$

Table 17: Basic likelihood geometry and calibration quantities for the diabetes application.

Quantity	Value
n	442
p	10
$\text{rank}(X)$	10
$\lambda_{\min}(X^\top X)$	0.00856053
$\lambda_{\max}(X^\top X)$	4.02421418
$\kappa(X^\top X)$	470.09
Maximum absolute predictor correlation	0.896663
Residual RSS	212.672503
Residual degrees of freedom	431
$\hat{\sigma}$	0.702453
$\hat{\sigma}^2$	0.493440
L_f	8.155433
$\lambda = 1/L_f$	0.122618
δ	0.055178
θ_0	0.222812
Frozen $\hat{\theta}$	0.337289

The final iterate is 0.344333. Mean values over the first and second halves of the averaging region are 0.338460 and 0.336157, respectively, a relative split difference of approximately -0.68% . Over the final 5000 iterations, the mean and standard deviation of the parameter path are 0.337614 and 0.003197. No parameter-bound hits occur, and the full SAPG audit takes approximately 6.68 seconds.

5.2 Nonsmoothed MAP and posterior-computation audits

Conditional on the frozen $(\hat{\sigma}^2, \hat{\theta})$, FISTA converges to the nonsmoothed MAP in 1412 iterations. Its support is

$$\{\text{Sex}, \text{BMI}, \text{BP}, \text{S1}, \text{S3}, \text{S4}, \text{S5}, \text{S6}\},$$

with Age and S2 exactly zero. The smoothed MAP is retained only as an approximation diagnostic. Table 18 records the coordinate-level comparison.

For this application we record the simple relative mode displacement

$$\frac{\|\hat{\beta}_{\text{MAP},\lambda} - \hat{\beta}_{\text{MAP}}\|_2}{\|\hat{\beta}_{\text{MAP}}\|_2} = 0.002023.$$

Thus the chosen Moreau scale changes the mode only very slightly; the nonsmoothed MAP remains the point estimate used by the decision rule.

The first plain-MYULA calculation used 4000 burn-in transitions and 4000 retained draws. It was useful for reconnaissance, but the weakest likelihood directions were insufficiently explored. For the reported calculation, the fitted model and numerical scales are fixed at their final values, and the short and extended summaries are obtained from the same scalar-step MYULA trajectory and random-number stream, initialised at the nonsmoothed

Table 18: Exact nonsmoothed and Moreau-smoothed MAPs at the diabetes calibration.

Variable	Nonsmoothed MAP	Smoothed MAP
Age	0.000000	−0.011144
Sex	−2.746641	−2.745396
BMI	6.804027	6.804206
BP	3.964745	3.967025
S1	−1.931291	−1.916566
S2	0.000000	−0.011019
S3	2.447705	2.452710
S4	0.680066	0.678736
S5	6.773984	6.769802
S6	0.772569	0.774246

MAP corresponding to the fixed $\hat{\theta}$. For the final analysis, the burn-in endpoint is moved to transition 20,000, after which 80,000 draws are retained; the long-run retained sample is therefore not the original 4000-draw sample with additional draws appended. Table 19 summarises the short-versus-long audit.

Table 19: Numerical-adequacy diagnostics for the initial and extended plain-MYULA diabetes calculations.

Diagnostic	Initial run	Extended run
Retained draws	4000	80000
Median coordinate ESS	88.38	1589.59
Minimum coordinate ESS	8.78	198.81
Weakest eigendirection ESS	8.48	192.23
Median eigendirection ESS	198.27	3321.10
Strongest eigendirection ESS	1152.59	22414.31
Potential ESS	31.96	2447.38
Median posterior SD	1.078680	1.109263
Posterior-mean half-split gap	0.0276	0.0547

Here “Potential ESS” is the ESS of the scalar sequence $U_{\hat{\theta},\lambda}(\beta^{(m)})$, and the posterior-mean half-split gap is the relative Euclidean displacement between posterior means computed from the first and second halves of the retained draws. The half-split gap is a descriptive stability check rather than a quantity expected to decrease monotonically with run length: the extended chain explores weak likelihood directions that were only sparsely visited in the short calculation, so its two retained halves can differ more even while the directional ESS improves substantially. The short-to-long posterior-mean relative displacement is 0.0769, and the largest change in a marginal sign probability is 0.0922. These diagnostics suggested that a longer run would be preferable: the complete 100,000-transition calculation takes approximately 20.9 seconds, so no preconditioner or Moreau retuning is needed.

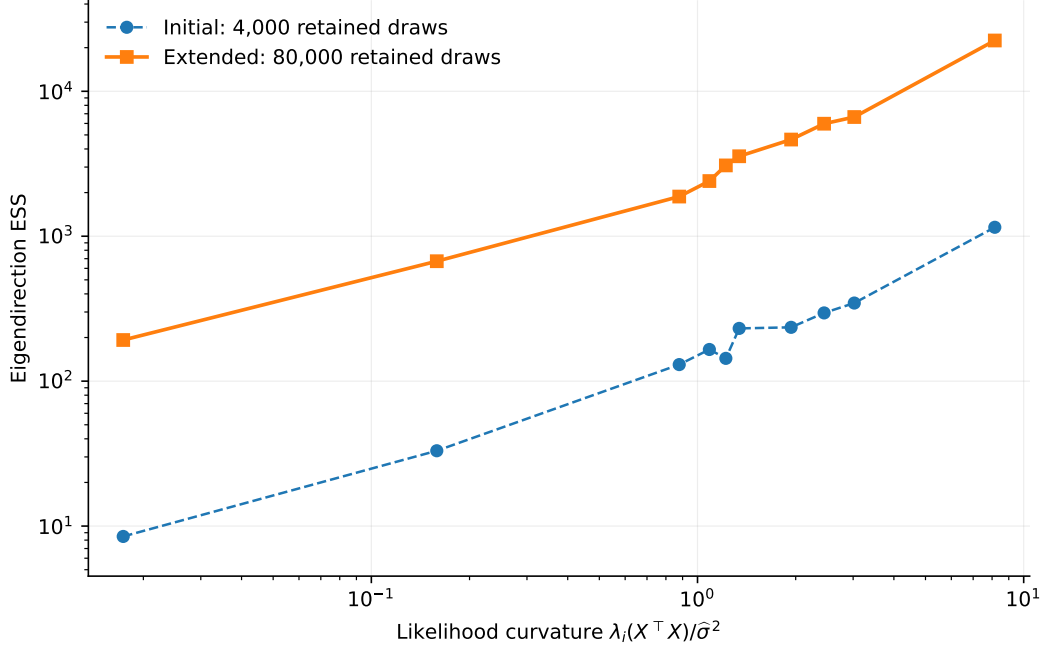


Figure 4: Eigendirection effective sample size (ESS) against likelihood curvature for the initial and extended scalar-step MYULA calculations on the diabetes data. Directions are eigenvectors of $X^T X$, and the horizontal axis shows $\lambda_i(X^T X)/\hat{\sigma}^2$; both axes are logarithmic. The initial calculation retained 4,000 draws and the extended calculation 80,000, with the fitted model and numerical scales held fixed. Mixing is slowest in the weakest likelihood directions, while the extended calculation raises the weakest-direction ESS from 8.48 to 192.23.

Table 20: Directional ESS before and after the long-run diabetes audit.

Direction	Eigenvalue $X^T X$	Likelihood curvature	Short ESS	Long ESS	ESS factor
1	0.008561	0.017349	8.48	192.23	22.67
2	0.078320	0.158722	33.09	672.21	20.31
3	0.433683	0.878898	130.10	1877.92	14.43
4	0.536560	1.087388	165.38	2402.55	14.53
5	0.602719	1.221465	143.62	3080.16	21.45
6	0.662186	1.341979	231.15	3562.03	15.41
7	0.955476	1.936359	235.14	4645.75	19.76
8	1.205962	2.443991	295.62	5974.26	20.21
9	1.492318	3.024317	345.85	6644.46	19.21
10	4.024214	8.155433	1152.59	22414.31	19.45

5.3 Coefficient summaries and decision diagnostics

Table 21 records the nonsmoothed MAP, long-chain posterior summaries, reference activation probabilities and reference selection indicators. The four marginal 95% credible intervals excluding zero are those for Sex, BMI, BP and S5, whereas the reference MAP-plus-posterior action selects BMI, BP and S5.

Table 21: Full coefficient-level summary for the frozen diabetes analysis. Activation probabilities use $k = 2.5$; the selected column additionally applies $\pi_\star = .75$.

Variable	MAP	Post. mean	Post. SD	2.5%	97.5%	Activation	Selected
Age	0.000000	-0.054562	0.727229	-1.498994	1.369171	0.000200	0
Sex	-2.746641	-2.781635	0.808899	-4.351513	-1.183748	0.503075	0
BMI	6.804027	6.804885	0.877289	5.082087	8.504506	1.000000	1
BP	3.964745	3.983405	0.850941	2.296089	5.632461	0.920975	1
S1	-1.931291	-2.542582	2.385527	-7.916032	1.653159	0.439150	0
S2	0.000000	0.151429	1.978284	-3.542756	4.569156	0.155562	0
S3	2.447705	1.902562	1.528850	-0.992071	4.926575	0.286950	0
S4	0.680066	1.326592	1.560312	-1.625660	4.521838	0.184025	0
S5	6.773984	6.840286	1.341236	4.272900	9.550659	0.999450	1
S6	0.772569	0.831751	0.822946	-0.709894	2.511301	0.011775	0

For reference, the OLS coefficients in the correlated serum block are strongly reallocated by the Laplace fit. In particular, OLS gives $(S1, S2) \approx (-10.28, 6.18)$, the nonsmoothed MAP gives approximately $(-1.93, 0)$, and the posterior means are approximately $(-2.54, 0.15)$. S3 also changes sign between OLS and the MAP/posterior summaries. These differences are not given a causal interpretation; they record the instability of coordinate attribution under the observed correlation structure.

The full gate diagnostics are shown in Table 22. Here $D_{\min, \text{elig}}$ is the smallest Monte Carlo decision distance among coefficients passing the MAP magnitude gate; this is the quantity most directly relevant to the two-stage sparse action.

At the reference operating point, the three MAP-eligible coordinates are BMI, BP and S5, and all three pass the posterior gate. The minimum eligible decision distance is 46.99, so the reference sparse action is not close to changing because of Monte Carlo uncertainty. Small values of $D_{\min, \text{all}}$ at other grid settings can instead be driven by coordinates that do not pass the MAP magnitude gate, and therefore need not indicate instability of the final selected support.

5.4 External benchmark details

The external comparison is reconstructed from numerical interval output recorded in the public notebook accompanying Zhou et al. [2024]; it is not an independent rerun of the benchmark MCMC algorithms. The Bayesian columns are ProxMCMC, Bayesian lasso with reversible-jump selection (RJ-T), Bayesian lasso without reversible-jump selection (RJ-F), the horseshoe, and the present EB-Laplace + MYULA analysis. The notebook additionally reports selective-inference intervals after a 10-fold cross-validated lasso using the one-

Table 22: Full diabetes decision-gate diagnostics. The table uses the same prespecified $(k, \pi_\star)$ grid as the main manuscript.

k	$\pi_\star$	Threshold	MAP-eligible size	Selected size	$D_{\min, \text{all}}$	$D_{\min, \text{elig}}$
2.0	0.50	2.218525	5	4	1.477	3.519
2.0	0.75	2.218525	5	4	1.555	1.555
2.0	0.90	2.218525	5	3	12.441	21.657
2.5	0.50	2.773157	3	3	0.406	115.700
2.5	0.75	2.773157	3	3	10.710	46.990
2.5	0.90	2.773157	3	3	5.765	5.765
3.0	0.50	3.327788	3	3	5.701	42.199
3.0	0.75	3.327788	3	3	4.531	4.531
3.0	0.90	3.327788	3	2	18.069	18.069

Table 23: Reference-setting activation probabilities and Monte Carlo decision distances for the diabetes analysis.

Variable	Activation prob.	MAP eligible	Selected	D_j
Age	0.000200	0	0	8486.55
Sex	0.503075	0	0	32.64
BMI	1.000000	1	1	∞
BP	0.920975	1	1	46.99
S1	0.439150	0	0	10.71
S2	0.155562	0	0	42.72
S3	0.286950	0	0	24.64
S4	0.184025	0	0	46.12
S5	0.999450	1	1	893.35
S6	0.011775	0	0	725.36

standard-error rule. Those intervals are conditional frequentist confidence intervals and are retained only as a descriptive comparison.

Table 24: Recorded 95% diabetes intervals used in the external benchmark reconstruction.

Variable	ProxMCMC	BL RJ-T	BL RJ-F	Horseshoe	SelInf	Present Method
Age	[-1.238,1.208]	[-.867,.804]	[-1.362,1.316]	[-.986,.866]	–	[-1.499,1.369]
Sex	[-3.881,-.730]	[-4.186,-.761]	[-4.181,-1.024]	[-3.936,-.213]	[-4.346,-1.762]	[-4.352,-1.184]
BMI	[5.111,8.450]	[5.197,8.547]	[5.040,8.474]	[5.309,8.811]	[5.399,8.188]	[5.082,8.505]
BP	[1.984,5.365]	[2.323,5.634]	[2.200,5.578]	[1.936,5.500]	[2.887,5.582]	[2.296,5.632]
S1	[-4.118,.986]	[-5.921,.441]	[-6.338,1.393]	[-4.606,.435]	–	[-7.916,1.653]
S2	[-2.791,1.626]	[-2.683,3.147]	[-3.295,3.377]	[-2.185,1.593]	–	[-3.543,4.569]
S3	[-.011,4.434]	[0,5.025]	[-.480,4.832]	[-.109,4.769]	[2.346,5.151]	[-.992,4.927]
S4	[-1.121,3.703]	[-1.258,4.246]	[-1.457,4.250]	[-1.139,3.309]	–	[-1.626,4.522]
S5	[4.172,8.452]	[4.541,9.156]	[4.316,9.085]	[4.643,9.077]	[4.755,7.562]	[4.273,9.551]
S6	[-.547,2.271]	[-.357,1.994]	[-.659,2.404]	[-.519,1.732]	–	[-.710,2.511]

The corresponding zero-exclusion pattern is shown in Table 25. All five Bayesian analyses give the same pattern: Sex, BMI, BP and S5 exclude zero, while the remaining six intervals contain zero. The selective-inference column additionally excludes zero for S3.

Table 25: Diabetes 95% interval zero-exclusion pattern. One indicates that the reported interval excludes zero; a dash means that selective inference reported no interval because the variable was outside the lasso-selected model.

	ProxMCMC	BL RJ-T	BL RJ-F	Horseshoe	SelInf	Present Method
Age	0	0	0	0	–	0
Sex	1	1	1	1	1	1
BMI	1	1	1	1	1	1
BP	1	1	1	1	1	1
S1	0	0	0	0	–	0
S2	0	0	0	0	–	0
S3	0	0	0	0	1	0
S4	0	0	0	0	–	0
S5	1	1	1	1	1	1
S6	0	0	0	0	–	0

For BL RJ-T, the recorded lower endpoint for S3 is numerically equal to zero. The zero-exclusion summary therefore treats that interval as containing zero, using the conservative rule “lower > 0 or upper < 0 ”. This convention matches the notebook-level audit.

The lasso-selected model underlying the selective-inference calculation is

$$\{\text{Sex, BMI, BP, S3, S5}\}.$$

For comparison, Table 26 records the number of prespecified decision-grid settings in which the present procedure selects each variable. The two additional selective-inference variables are exactly the two boundary cases emphasised in the main manuscript: Sex is directionally well identified but lies near the practical-magnitude threshold, while S3 can be MAP-eligible

Table 26: Descriptive comparison of the present diabetes decision-support summary with the lasso support used for selective inference. “Grid count” is the number of the nine prespecified $(k, \pi_\star)$ settings in which the present method selects the variable.

Variable	Reference activation	Grid count	SelInf selected
Age	0.0002	0/9	No
Sex	0.5031	2/9	Yes
BMI	1.0000	9/9	Yes
BP	0.9210	8/9	Yes
S1	0.4392	0/9	No
S2	0.1556	0/9	No
S3	0.2870	0/9	Yes
S4	0.1840	0/9	No
S5	0.9995	9/9	Yes
S6	0.0118	0/9	No

under the permissive magnitude threshold without receiving sufficient posterior activation support.

The support comparison is descriptive only. The selective-inference lasso support and the present MAP-plus-posterior action arise from different procedures, and the selective-inference intervals are not Bayesian credible intervals.

6 Intrinsic-dimensional calibration under a hard homogeneous constraint

This supplementary experiment provides a paired numerical verification of the intrinsic-dimensional calibration derived in the main manuscript for the hard homogeneous constraint

$$C_0 = \{\beta \in \mathbb{R}^p : \mathbf{1}^\top \beta = 0\}.$$

The comparator deliberately replaces the intrinsic dimension by the ambient one and is used only as a diagnostic, not as an implementation alternative. The main manuscript reports the resulting calibration shift in summary; here we record the exact simulation design, paired score-root check, and downstream MAP, posterior and support comparisons.

6.1 Design and paired comparison

We use $R = 100$ independent datasets with $n = 200$ and $p = 10$. The true coefficient vector is

$$\beta^\star = (1, -1, 0.8, -0.8, 0, \dots, 0)^\top, \quad \mathbf{1}^\top \beta^\star = 0,$$

and the observation noise is Gaussian with known $\sigma = 1$. Each design matrix is generated with i.i.d. standard Gaussian entries and then each realised column is rescaled to Euclidean norm $\sqrt{n}$. The resulting likelihood geometry is intentionally benign: across the ensemble,

the mean restricted condition number is 1.9329 and the mean empirical signal-to-noise ratio is 3.3066. The purpose is to isolate the intrinsic-dimensional calibration effect rather than confound it with multicollinearity or an ill-conditioned likelihood.

Let $d = p - 1 = 9$. We represent the constrained parameter as $\beta = Qz$, where $Q \in \mathbb{R}^{p \times d}$ has orthonormal columns spanning C_0 . Constrained least squares is then obtained from

$$\hat{z}_{\text{CLS}} \in \arg \min_{z \in \mathbb{R}^d} \|y - XQz\|_2^2, \quad \beta_{\text{CLS}} = Q\hat{z}_{\text{CLS}},$$

so that $\mathbf{1}^\top \beta_{\text{CLS}} = 0$ by construction. Both SAPG branches use the common initialisation

$$\theta_0 = \frac{d}{\|\beta_{\text{CLS}}\|_1}.$$

The intrinsic-dimensional branch uses score constant $d = 9$, whereas the ambient-dimensional comparator uses $p = 10$. Thus the intended experimental perturbation is the dimension entering the SAPG score; the data, initial chain state, numerical scales, stochastic-approximation schedule and innovation streams are otherwise paired.

The restricted likelihood curvature is

$$L_{f,C_0} = \frac{\|XQ\|_2^2}{\sigma^2},$$

and we use

$$\lambda = L_{f,C_0}^{-1}, \quad \delta = \frac{0.9}{L_{f,C_0} + 1/\lambda} = \frac{0.9}{2L_{f,C_0}}.$$

SAPG is run for 15,000 iterations, with one constrained-MYULA transition per stochastic-approximation step. The reported EB estimate is obtained by averaging the SAPG iterates on the log scale, $\eta = \log \theta$, after discarding the first 4,000 stochastic-approximation updates, and then exponentiating the resulting tail average. We use a deliberately wide projection interval, $[\theta_0/10^6, 10^6\theta_0]$, chosen so that projection does not constrain the iterates; it is inactive in all runs. After fixing the two EB estimates, we compute the original nonsmoothed constrained MAP under each branch. Final posterior calculations use constrained MYULA with 4,000 burn-in iterations and 4,000 retained draws, with common random numbers across the paired posterior chains. The decision rule is evaluated on

$$k \in \{2, 2.5, 3\}, \quad \pi_\star \in \{0.5, 0.75, 0.9\},$$

with (2.5, 0.75) used as the reference setting.

6.2 Calibration result and moment check

The dimensional benchmark is

$$\frac{p}{d} = \frac{10}{9} = 1.111111.$$

Every replicate satisfies $\hat{\theta}_p > \hat{\theta}_d$, and every paired ratio also exceeds 10/9. At the two score roots,

$$d = \theta_d m(\theta_d), \quad p = \theta_p m(\theta_p), \quad m(\theta) = \mathbb{E}_\theta[\|\beta\|_1 \mid y],$$

Table 27: Intrinsic-dimensional SAPG calibration over $R = 100$ paired replicates.

Quantity	Mean	SD	Median	Min	Max
$\hat{\theta}_d$	2.228029	.082104	2.226848	2.063264	2.450133
$\hat{\theta}_p$	2.480854	.091691	2.479560	2.296486	2.728680
$\hat{\theta}_p/\hat{\theta}_d$	1.113471	.000281	1.113414	1.112760	1.114246
Ratio predicted from score-root moments	1.113469	.000278	1.113416	1.112779	1.114218

so that

$$\frac{\theta_p}{\theta_d} = \frac{p m(\theta_d)}{d m(\theta_p)}.$$

Using the posterior moment estimates from the paired runs gives a mean predicted ratio of 1.113469. The mean absolute relative discrepancy between the observed and predicted ratios is 1.94×10^{-5} , with a maximum 6.73×10^{-5} . The SAPG diagnostics show no boundary hits; median absolute tail slopes of $\eta = \log \theta$ are 2.04×10^{-7} and 2.00×10^{-7} for the intrinsic- and ambient-dimensional branches, respectively. As a direct check that the constrained chain remains in the sum-zero subspace, we monitor the absolute violation $|\mathbf{1}^\top \beta^{(k)}|$ along the SAPG/MYULA states used for calibration. Its maximum over all paired runs is 1.62×10^{-15} , i.e. at floating-point scale.

6.3 Downstream MAP and posterior consequences

In this subsection, the within-branch Moreau-MAP shift is the Euclidean distance

$$\left\| \hat{\beta}_{\text{MAP},\lambda} - \hat{\beta}_{\text{MAP}} \right\|_2$$

computed at that branch’s fitted $\hat{\theta}$.

The larger ambient-dimensional EB scale produces slightly stronger shrinkage in every replicate: the ratio of ambient-to-intrinsic MAP ℓ_1 norms has mean 0.996888, and is below one in all 100 datasets. The absolute coefficient displacement nevertheless remains small.

Table 28: Downstream consequences of the intrinsic-dimension calibration perturbation. Error and risk differences are ambient-dimensional minus intrinsic-dimensional, so negative values favour the ambient-dimensional branch under that particular finite-sample loss.

Quantity	Ensemble summary	Additional diagnostic
MAP ℓ_1 ratio, ambient/intrinsic	mean .996888	ambient smaller in 100%
MAP branch displacement $\ \hat{\beta}_p - \hat{\beta}_d\ _2$	mean 4.046×10^{-3}	median 4.033×10^{-3}
Relative- L_2 error difference	mean -9.573×10^{-4}	ambient lower in 93%
True-support RMSE difference	mean $+1.731 \times 10^{-5}$	ambient lower in 49%
Prediction-risk difference	mean -6.285×10^{-4}	ambient lower in 96%
Median $\ \hat{\beta}_{\text{MAP},\lambda} - \hat{\beta}_{\text{MAP}}\ _2$, intrinsic	2.134×10^{-3}	exceeds branch MAP shift in 23%
Median $\ \hat{\beta}_{\text{MAP},\lambda} - \hat{\beta}_{\text{MAP}}\ _2$, ambient	2.842×10^{-3}	exceeds branch MAP shift in 35%
Median posterior-SD ratio, ambient/intrinsic	.997348	mean .997340
Posterior-mean branch displacement	mean 3.158×10^{-3}	median 3.134×10^{-3}
Posterior-SD branch displacement	mean 7.036×10^{-4}	–

The median MAP branch displacement is about 0.22% of $\|\beta^*\|_2$. The point-estimation comparisons also show that the intrinsic-dimensional calibration implied by the stated constrained prior need not coincide with the branch that minimises a particular finite-sample loss: the more strongly shrunk ambient-dimensional branch slightly improves overall relative- L_2 error and prediction risk in most replications, principally by damping null-coordinate noise, even though it is not the calibration implied by the constrained prior. By contrast, RMSE restricted to the true support is effectively tied.

Posterior consequences are comparably small in this experiment. The mean Euclidean displacement between posterior means is 3.16×10^{-3} , while the median ratio of marginal posterior scales is 0.99735. These small branch differences are nevertheless well resolved relative to their Monte Carlo uncertainty: median coefficient ESS is 781.4 for the intrinsic-dimensional branch and 788.4 for the ambient-dimensional branch, and the median MCSE of coordinatewise paired posterior-mean differences is 2.216×10^{-5} . This MCSE is computed from the paired-difference sequence $\beta_{p,j}^{(m)} - \beta_{d,j}^{(m)}$, using the common-random-number coupling of the two posterior chains. The conclusion is that the posterior differences are substantively small, but are not explained by poorly resolved chains.

6.4 Support decisions and computational checks

At the reference setting $(k, \pi_\star) = (2.5, 0.75)$, both branches recover the exact four-variable support in every replicate. The complete grid is shown in Table 29. The branches select identical supports everywhere except at $(k, \pi_\star) = (2, 0.5)$, where they differ in one of the 100 datasets. Thus the substantial EB-scale shift does not materially alter support recovery in this deliberately well-identified example.

Table 29: Posterior-informed support recovery over the full intrinsic-dimension verification grid. “Same” stands for the fraction of replicates for which the intrinsic- and ambient-dimensional branches select exactly the same support.

k	$\pi_\star$	TPR_d	TPR_p	FDR_d	FDR_p	$ \hat{S}_d $	$ \hat{S}_p $	Exact_d	Exact_p	Same
2.0	.50	1.00	1.00	.0273	.0260	4.14	4.13	.87	.87	.99
2.0	.75	1.00	1.00	0	0	4.00	4.00	1.00	1.00	1.00
2.0	.90	1.00	1.00	0	0	4.00	4.00	1.00	1.00	1.00
2.5	.50	1.00	1.00	.0020	.0020	4.01	4.01	.99	.99	1.00
2.5	.75	1.00	1.00	0	0	4.00	4.00	1.00	1.00	1.00
2.5	.90	1.00	1.00	0	0	4.00	4.00	1.00	1.00	1.00
3.0	.50	1.00	1.00	0	0	4.00	4.00	1.00	1.00	1.00
3.0	.75	1.00	1.00	0	0	4.00	4.00	1.00	1.00	1.00
3.0	.90	1.00	1.00	0	0	4.00	4.00	1.00	1.00	1.00

The numerical checks confirm that the paired comparison is not being driven by unstable optimisation or sampling: the hard sum-zero relation is preserved to floating-point accuracy throughout, the SAPG projection bounds are inactive, and the posterior ESS and paired-difference MCSE diagnostics indicate that the reported branch differences are well resolved numerically. The main conclusion is therefore deliberately narrow: using the ambient dimen-

Table 30: Selected numerical and Monte Carlo diagnostics for the intrinsic-dimension verification.

Diagnostic	Value
Mean empirical SNR	3.3066
Mean restricted condition number	1.9329
Mean $L_{f,C_0}/L_f$	.980119
Maximum SAPG boundary-hit fraction	0
Maximum SAPG constraint residual	1.62×10^{-15}
Median $ \eta $ tail slope, intrinsic branch	2.04×10^{-7}
Median $ \eta $ tail slope, ambient branch	2.00×10^{-7}
Maximum MAP/smoothed-MAP constraint residual	3.44×10^{-12}
Median coefficient ESS, intrinsic posterior	781.4
Median coefficient ESS, ambient posterior	788.4
Worst minimum coordinate ESS, intrinsic	433.1
Worst minimum coordinate ESS, ambient	433.9
Median paired-difference MCSE	2.216×10^{-5}
Worst paired-difference MCSE	4.742×10^{-5}
Maximum posterior constraint residual	4.58×10^{-16}

sion in place of the intrinsic dimension produces a large, reproducible shift in the EB scale, while this particular well-conditioned problem is insensitive to that perturbation at the final support-selection stage.

7 Additional sensitivity analysis: observation scale and Moreau parameter

This additional study uses a single fixed unconstrained synthetic dataset to examine propagation through the fitted workflow without introducing dataset-to-dataset variation. It complements Sections 2.1 and 2.2 of the main manuscript by perturbing, separately, the supplied observation variance and the Moreau smoothing parameter. The fixed dataset uses the same unconstrained synthetic design and coefficient pattern as Study 2 at $\kappa_c = 0$: $n = 200$, $p = 100$,

$$\beta^* = (1, 0.8, 0.6, 0.4, -0.8, -0.5, -0.3, -0.2, 0, \dots, 0)^\top,$$

the entries of X are generated independently from $N(0, 1)$ and each realised column is rescaled to Euclidean norm $\sqrt{n}$, and $y = X\beta^* + \varepsilon$ with $\varepsilon \sim N(0, I_n)$, so the true $\sigma = 1$. The same realised X and y are used throughout both sensitivity experiments. The baseline residual-variance estimate is

$$\hat{\sigma}_0^2 = 0.941247,$$

and, writing

$$L_{f,0} = \frac{\lambda_{\max}(X^\top X)}{\hat{\sigma}_0^2},$$

the baseline Moreau parameter is

$$\lambda_0 = L_{f,0}^{-1}.$$

The reference sparse decision uses $(k, \pi_\star) = (2.5, 0.75)$. In the posterior-computation summaries below, Weak ESS denotes the effective sample size after projection onto the weakest likelihood eigendirection, and D_{elig} denotes the smallest decision distance D_j among MAP-eligible coordinates.

As a finite-sample motivation for the observation-variance experiment, we also compute a robust residual scale on this same fixed dataset. Let r_i denote the OLS residual and h_{ii} the corresponding leverage. Since $\text{Var}(r_i | X) = \sigma^2(1 - h_{ii})$, we first standardise

$$\tilde{r}_i = \frac{r_i}{\sqrt{1 - h_{ii}}},$$

and then use the normal-consistent MAD

$$\hat{\sigma}_{\text{MAD,lev}} = \frac{\text{median}_i |\tilde{r}_i - \text{median}_\ell \tilde{r}_\ell|}{\Phi^{-1}(0.75)}.$$

For this realisation,

$$\hat{\sigma}_{\text{MAD,lev}} = 0.901285, \quad \hat{\sigma}_{\text{MAD,lev}}^2 = 0.812314,$$

whereas the classical residual calculation gives $\hat{\sigma}_0 = 0.970179$ and $\hat{\sigma}_0^2 = 0.941247$. Thus the robust estimate corresponds to

$$c_{\text{MAD}} = \frac{\hat{\sigma}_{\text{MAD,lev}}^2}{\hat{\sigma}_0^2} = 0.863019, \quad \frac{L_f(c_{\text{MAD}})}{L_{f,0}} = 1.158724.$$

This calculation is not intended as an estimator comparison or as evidence that MAD should replace the classical residual estimator. Indeed, in this synthetic realisation the latter is closer to the true $\sigma = 1$. Its role is simply to show that two defensible preliminary plug-in scales can differ non-negligibly in finite samples, providing an empirical anchor for the controlled perturbation below.

The two experiments perturb different objects. The observation-variance experiment changes the statistical target itself by replacing $\hat{\sigma}_0^2$ with a supplied value σ_c^2 . The Moreau experiment instead fixes the supplied variance and changes the smoothing parameter used in the SAPG inner sampler and final posterior calculation. In both experiments, SAPG is rerun after the perturbation, so any induced change in $\hat{\theta}$ is propagated through the nonsmoothed MAP and posterior decision.

7.1 Observation-variance perturbation

We set

$$\sigma_c^2 = c_\sigma \hat{\sigma}_0^2, \quad c_\sigma \in \{2, 1, 0.5, 0.1, 0.01\},$$

and keep the Moreau parameter fixed at λ_0 . Because

$$L_f(c_\sigma) = \frac{\lambda_{\max}(X^\top X)}{\sigma_c^2} = \frac{L_{f,0}}{c_\sigma},$$

the likelihood curvature varies inversely with the supplied variance even though the Moreau scale is held fixed. The MYULA timestep is therefore recomputed from the branch-specific Lipschitz bound.

For the nonsmoothed MAP,

$$\frac{1}{2\sigma_c^2} \|y - X\beta\|_2^2 + \hat{\theta} \|\beta\|_1,$$

multiplication by σ_c^2 shows that the relative penalty scale is $\sigma_c^2 \hat{\theta}$. Table 31 records the resulting propagation. The empirical Bayes estimate changes comparatively little, whereas the effective penalty collapses under severe downward variance misspecification. Posterior marginal scales and hence $\tau_{\text{post}} = 2.5 \text{ median}_j \widehat{\text{sd}}(\beta_j \mid y)$ contract at the same time.

Table 31: Observation-variance sensitivity on the fixed sensitivity-analysis dataset. Selection columns use the reference decision setting.

c_σ	σ_c^2	$L_f/L_{f,0}$	$\hat{\theta}$	$\sigma_c^2 \hat{\theta}$	MAP rel. L_2	Med. SD	τ_{post}	Size	TPR	FDR
2	1.882494	0.5	9.496410	17.8769	.220675	.082637	.206591	6	.750	.000000
1	.941247	1	9.590359	9.0269	.257992	.068221	.170551	7	.875	.000000
.5	.470624	2	9.490772	4.4666	.329051	.055677	.139193	8	.875	.125000
.1	.094125	10	8.769990	.8255	.475577	.030101	.075253	34	1.000	.764706
.01	.009412	100	8.354502	.07864	.529882	.009994	.024986	79	1.000	.898734

The apparent gain in TPR at the two smallest supplied variances is therefore not a useful recovery improvement: all eight true coordinates are selected, but they are accompanied by 26 and 71 false positives at the reference gate. The practical threshold falls by almost an order of magnitude between $c_\sigma = 1$ and 0.01, while the effective penalty relative to the least-squares term falls by more than two orders of magnitude.

We additionally passed the MAD-implied branch $c_{\text{MAD}} = 0.863019$ through exactly the same SAPG–FISTA–MYULA workflow, with the absolute Moreau parameter fixed at λ_0 . The empirical Bayes estimate is essentially unchanged, 9.589375 versus 9.590359 at baseline, but the effective penalty $\sigma^2 \hat{\theta}$ decreases from 9.0269 to 7.7896. Median posterior SD decreases from 0.06822 to 0.06535, and the reference practical threshold from 0.17055 to 0.16338. Nevertheless, the reference decision is identical: both branches select the same seven true coordinates, with TPR 0.875 and FDR zero. Across the complete 3×3 gate grid, the selected supports agree in eight of the nine settings; the only difference occurs at $(k, \pi_\star) = (3, 0.9)$, where the MAD branch retains one additional true coordinate. Thus the estimator-induced discrepancy visibly propagates through the fitted posterior without changing the principal sparse conclusion. The wider sweep in Table 31 should therefore be read as a controlled stress test of how that practical stability can fail.

For the support comparisons below, if $\hat{S}_a$ and $\hat{S}_b$ are two selected sets, their symmetric difference is

$$\hat{S}_a \triangle \hat{S}_b = (\hat{S}_a \setminus \hat{S}_b) \cup (\hat{S}_b \setminus \hat{S}_a),$$

so that $|\hat{S}_a \triangle \hat{S}_b|$ counts coordinates selected in exactly one of the two calculations. We also report the Jaccard index

$$J(\hat{S}_a, \hat{S}_b) = \frac{|\hat{S}_a \cap \hat{S}_b|}{|\hat{S}_a \cup \hat{S}_b|},$$

for which $J = 1$ corresponds to identical nonempty supports and smaller values indicate increasing disagreement.

Matched-total-time posterior audits. The severe branches also have much smaller stable MYULA timesteps, so their initial fixed-transition calculations cover less Langevin time than the baseline run. We therefore reran the $c_\sigma = 0.1$ and 0.01 branches with the total simulated Langevin time,

$$T \approx N_{\text{step}}\delta,$$

matched to the baseline value $T = 6.121852$. This required 44,000 and 404,000 total transitions, respectively. Table 32 compares the initial and matched-time calculations.

Table 32: Matched-total-Langevin-time audits for the two severe downward variance perturbations.

c_σ	Total steps	Med. ESS short	Med. ESS audit	Weak ESS short	Weak ESS audit	D_{elig} short	D_{elig} audit	Size short/audit	FDR short/audit
.1	44,000	211.2	1144.1	87.4	400.2	.0369	.9409	34/34	.7647/.7647
.01	404,000	184.4	3634.9	62.7	2600.0	.1743	3.2496	79/78	.8987/.8974

The audits greatly improve ESS and move the decision-distance diagnostic away from the probability boundary, but do not repair the sparse decision. The $c_\sigma = 0.1$ support is identical before and after the audit. At $c_\sigma = 0.01$, only one coordinate changes and the support Jaccard index is 0.9873. Posterior relative L_2 error and median marginal SD are likewise essentially unchanged. These checks distinguish a Monte Carlo precision problem from a target-calibration problem: the latter is responsible for the large false discovery rates.

7.2 Moreau-parameter perturbation

The second sensitivity experiment keeps

$$\sigma^2 = \hat{\sigma}_0^2$$

fixed and varies

$$r_\lambda = \frac{\lambda}{\lambda_0} \in \{0.25, 0.5, 1, 2, 4\}.$$

SAPG is rerun at every value because its stochastic score is estimated using a MYULA kernel and can therefore inherit numerical dependence on the smoothing scale. The exact nonsmoothed MAP is then recomputed at the resulting branch-specific $\hat{\theta}$, followed by the corresponding smoothed posterior calculation.

To separate the two propagation routes, we record two MAP diagnostics. Δ_{calib} is the relative L_2 displacement of the branch-specific nonsmoothed MAP from the baseline nonsmoothed MAP, so it measures propagation through the change in $\hat{\theta}$. Δ_{smooth} is the relative L_2 displacement between the smoothed and nonsmoothed MAPs within the same branch, and therefore directly tracks the Moreau approximation. Table 33 shows that the calibration component remains small over the whole sweep, while the direct smoothing discrepancy grows steadily as λ increases.

More precisely, writing $\hat{\beta}_{\text{MAP}}[\hat{\theta}(\lambda)]$ for the nonsmoothed MAP computed at the branch-specific empirical Bayes estimate, the two quantities reported in Table 33 are

$$\Delta_{\text{calib}}(\lambda) = \frac{\left\| \hat{\beta}_{\text{MAP}}[\hat{\theta}(\lambda)] - \hat{\beta}_{\text{MAP}}[\hat{\theta}(\lambda_0)] \right\|_2}{\left\| \hat{\beta}_{\text{MAP}}[\hat{\theta}(\lambda_0)] \right\|_2},$$

and

$$\Delta_{\text{smooth}}(\lambda) = \frac{\left\| \hat{\beta}_{\text{MAP}}^{\lambda}[\hat{\theta}(\lambda)] - \hat{\beta}_{\text{MAP}}[\hat{\theta}(\lambda)] \right\|_2}{\left\| \hat{\beta}_{\text{MAP}}[\hat{\theta}(\lambda)] \right\|_2}.$$

Thus the first normalises the calibration-induced movement by the norm of the baseline non-smoothed MAP, while the second normalises the direct smoothing distortion by the norm of the corresponding branch-specific nonsmoothed MAP. These are the branch-comparison normalisations used specifically in Table 33; they are distinct from the generic $D_2(\lambda)$ diagnostic introduced in the main text.

Table 33: Moreau-parameter sensitivity: empirical Bayes, approximation and estimation summaries.

λ/λ_0	$\hat{\theta}$	$\lambda\hat{\theta}$	Δ_{calib}	Δ_{smooth}	MAP rel. L_2	Post. mean rel. L_2	Med. SD
.25	9.896243	.004207	.006274	.010517	.255324	.303116	.064785
.50	9.761137	.008299	.003504	.019194	.256490	.305025	.066320
1	9.590359	.016309	.000000	.033354	.257992	.308166	.068221
2	9.400048	.031970	.003937	.056175	.259642	.312660	.070366
4	9.157094	.062287	.008969	.087811	.261815	.321499	.073278

The posterior computation becomes more efficient as λ increases and the Moreau approximation is more strongly smoothed. At the same time, the practical threshold increases because the posterior marginal scales increase. Despite these changes, the reference selected support is identical in all five branches.

Table 34: Moreau-parameter sensitivity: posterior-computation and reference-decision summaries.

λ/λ_0	δ	Med. ESS	Weak ESS	τ_{post}	Size	TPR	FDR	D_{elig}
.25	.000306	113.0	63.2	.161962	7	.875	0	16.57
.50	.000510	188.7	118.2	.165800	7	.875	0	16.61
1	.000765	297.6	167.5	.170551	7	.875	0	15.75
2	.001020	404.1	213.9	.175916	7	.875	0	15.37
4	.001224	464.7	240.8	.183196	7	.875	0	13.20

The selected support has Jaccard index one relative to the baseline branch for every value in the sweep. The minimum decision distance among MAP-eligible coordinates also remains

above 13 MCSE units at the reference operating point, so the observed support stability is not a consequence of probability estimates sitting close to the decision boundary. In this fixed dataset, the main effect of changing λ over the tested range is therefore a trade-off between approximation and posterior-sampling efficiency rather than a change in the final sparse action.

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